The empirical recurrence for OEIS A269106, proved by transfer matrix

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Abstract

OEIS A269106 counts arrays of width 5 over $\{0, 1, 2, 3\}$ in which the number of adjacent PAIRS of cells whose entries add up to 3 is at most one. The entry carries an empirical recurrence of order 14 contributed by R. H. Hardin. It is true, and it is decidable rather than empirical. The condition is global — no window of the array decides it — but the quantity it constrains is a count that may be capped at two, and a capped count is a bounded state; the result is a walk count on $S = 2105$ states.

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1 The conjecture

OEIS A269106 is “Number of n X 5 0..3 arrays with some element plus some horizontally or vertically adjacent neighbor totalling three no more than once.”. It has offset 1 and begins

$$756, 132300, 17593092, 2085484068, 232068730044, 24808345933548, 2579398703502996, 262780733311913580, \ldots$$

The entry states, as a conjecture contributed by its author and never marked settled:

> Empirical: a(n) = 236*a(n-1) -19546*a(n-2) +707906*a(n-3) -13261621*a(n-4) +137719076*a(n-5) -784626841*a(n-6) +2025590366*a(n-7) +1091882414*a(n-8) -18874661724*a(n-9) +33829878231*a(n-10) +10605114324*a(n-11) -91783964484*a(n-12) +94503314304*a(n-13) -30762353664*a(n-14) for n>15.

As of the “Last modified” line on the live entry (May 25 2024 21:15:23, revision 6) this is still recorded as empirical, and nothing on the entry records it as proved.

2 What is being counted

The array has 5 cells across, with entries in $\{0, 1, 2, 3\}$. An ADJACENT PAIR is a pair of cells

(i,j) and (i,j+1)
(i,j) and (i+1,j)

both lying inside the array. A pair is a pair of CELLS, so it is counted once and not once from each end. Writing k for the number of adjacent pairs whose two entries add up to 3, the entry counts the arrays with k at most one.

3 A capped count is a state

No window of the array decides the condition: k is a total over the whole array. But the condition distinguishes only $k = 0$, $k = 1$ and $k \geq 2$, so the count may be carried capped at two and stays bounded.

Lemma 1. *Let k_t be the number of qualifying pairs lying inside the first t rows. Then*

$$k_t \;=\; k_{t-1} + \big(\textit{pairs inside row } t\big) + \big(\textit{pairs joining rows } t-1 \textit{ and } t\big),$$

and both added terms are determined by rows $t-1$ and t alone.

Proof. Every adjacent pair lies either inside one row or across two consecutive rows, by the list above; a pair inside rows $1..t$ that is not inside rows $1..t-1$ must meet row t, so it is one of the two kinds counted. □

So the state is the pair $(\text{previous row}, \min(k, 2))$: a step appends a row, adds the pairs Lemma 1 names and caps the total at two. The end vector accepts the states with $k \leq 1$, so it is not the all-ones vector. With M the adjacency matrix,

$$a(n) \;=\; \iota^\top M^{\,j} \tau, \qquad S = 2105.$$

The walk index j and the entry's own n differ by a constant, read off by matching the published terms rather than assumed. In particular a is C-finite.

4 The proof

Lemma 2. *Let $q(t) = t^{14} - \sum_i c_i t^{14-i}$ be the characteristic polynomial of the conjectured recurrence. The residual $a(n) - \sum_i c_i a(n-i)$ equals $\iota^\top M^{\,j} q(M)\tau$ for the corresponding walk index j, so the recurrence holds from some point on if and only if $\iota^\top M^j q(M)\tau = 0$ for all large j, and it is enough to examine $j < S$.*

Proof. Factor M^j out of $M^{j+14} - \sum_i c_i M^{j+14-i}$. For the bound, M^S is an integer combination of $I, M, \ldots, M^{S-1}$ by Cayley–Hamilton, so the numbers $u_j = \iota^\top M^j q(M)\tau$ obey the monic recurrence given by the characteristic polynomial of M; S consecutive zeros force every later one, and the last nonzero u_j pins the threshold exactly. □

Theorem 3.

$$a(n) \;=\; 236\,a(n-1) - 19546\,a(n-2) + 707906\,a(n-3) - 13261621\,a(n-4) + 137719076\,a(n-5) - 784626841\,a(n-6)$$

for every $n > 15$.

Proof. The vector $w = q(M)\tau$ was formed by 14 matrix–vector products in exact integer arithmetic and $\iota^\top M^j w$ evaluated for $j = 0, 1, \ldots,$ again exactly; those integers vanish from the index corresponding to $n = 15 + 1$ onwards and the run continued until the working vector was identically zero. □

5 Verification

Four checks, each independent of the algebra above.

First, the model reproduces all 11 terms the entry publishes, exactly, in integer arithmetic.

Second, the count was recomputed for the smallest arrays straight from the definition: enumerate every array of that size in the orientation the entry's own name writes, count the adjacent pairs summing to 3 by running over the array once for each kind of pair, and keep the array when that count is at most one. No cap, no state and no transfer matrix are involved, and the published terms came back.

Third, the conjectured recurrence was evaluated directly on the published terms, with no matrices involved, and holds wherever the proved range and the data overlap.

Fourth, the annihilation test of Lemma 2 was carried out in exact integer arithmetic on the whole vector rather than on a sampled prefix.

References

[1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, https://oeis.org, 2026. Sequence A269106.

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[3] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.